

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

School of Computing — Department of Computer Science & Engineering

Student Name: Sautrik Roy	Reg No: RA2511003010052
Course: 21CSC203P — Advanced Programming Practice	Department: Computer Science and Engineering

Unit III | Session 2 | SLO 1: Thread Synchronization & Locks (Code: 3021)

Q: 1. Fill in the blank:

The _____ keyword in Java prevents concurrent access to a shared method or block.

Ans: Answer: synchronized

Q: 2. True or False:

Every Java object has an associated intrinsic lock (monitor).

Ans: Answer: True

Q: 3. Multiple Choice:

Which of the following guarantees atomic compound operations?

A) AtomicInteger B) int C) double D) boolean

Ans: Answer: A) AtomicInteger

Q: 4. Fill in the blank:

A thread acquires an _____ lock when entering a synchronized block.

Ans: Answer: intrinsic (or monitor)

Q: 5. What problem occurs if two threads increment an unsynchronized shared counter simultaneously?

Ans: Answer: Race Condition (Lost Update anomaly).

Q: 6. True or False:

Static synchronized methods lock on the Class object.

Ans: Answer: True

Q: 7. Fill in the blank:

The _____ class in `java.util.concurrent.locks` provides explicit locking mechanisms.

Ans: Answer: ReentrantLock

Q: 8. Fill in the blank:

The _____ keyword ensures visibility of changes to variables across threads.

Ans: Answer: volatile

Q: 9. Identify the issue with: `synchronized(new Object()) { ... }`

Ans: Answer: Ineffective locking because every thread synchronizes on a brand new object instance.

Q: 10. Fill in the blank:

ReentrantLock allows a thread to re-acquire a lock it already holds without causing _____.

Ans: Answer: deadlock